Lindsey Ma

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Education

Columbia University, M.S. in Statistics, GPA 3.71 *Sep 2024 – May 2026*

Coursework: Deep Learning, Advanced Machine Learning, Statistical Analysis of Neural Data, Robot Learning

Renmin University of China, B.A. in Economics, GPA 3.82 *Sep 2019 – Jun 2023*

Experiences

Research Engineer Intern - Applied ML for Plant Identification *May 2026 – Present*

Arborphy AI LLC *New York*

- Independently designed and implemented an EfficientNet–BioCLIP dual-encoder architecture with bidirectional spatial cross-attention, auxiliary organ supervision, and BioCLIP-anchored gated residuals to investigate feature complementarity and mitigate negative transfer; improved fusion macro-F1 from 0.808 to 0.832 across architecture iterations.
- Investigated semantic and geometric properties of BioCLIP, DINOv3, and EfficientNet representations using frozen linear probes, multi-seed clustering, nearest-neighbor retrieval, local PCA rotations, and centroid shrinkage to distinguish species identity, organ morphology, and photographic-scale confounding.
- Designed controlled fusion experiments comparing single encoders, feature concatenation, and unidirectional versus bidirectional cross-attention; conducted individual-disjoint, multi-seed, and grouped held-out evaluations, finding no reliable species-recognition advantage over BioCLIP alone.
- Developed individual- and species-disjoint benchmarks for species identification, organ classification, and cross-species retrieval; frozen DINOv3 features achieved 87.3% organ classification accuracy and 84.3% cross-species retrieval P@1. Analyzed cross-model errors and developed a 100-image human-review set spanning 85 species.
- Built a multi-agent botanical reasoning prototype with autonomous tool generation, specialized visual-analysis agents, constrained iterative planning, per-step information-gain evaluation, and knowledge-graph-based evidence aggregation; developed reproducible research pipelines and interactive analysis tools.

Software Developer - Real-Time Vision for Robotics *May 2025 – Dec 2025*

InnoX Academy Incubator Robotics Lab *Shenzhen*

- Developed a real-time vision pipeline for an autonomous badminton-training robot, integrating YOLOv8-based player detection, pose estimation, shuttlecock tracking, and court localization, achieving 20+ FPS.
- Implemented stereo 3D shuttlecock tracking and landing-point prediction; optimized Raspberry Pi inference with ONNX Runtime, quantization, and asynchronous video pipelines.
- Integrated a locally hosted language model through Ollama for voice-based robot commands and natural-language interaction, connecting user instructions with the robot's control workflow.

Research

Research Assistant - Self-Supervised Learning for Neural Imaging *Jan 2026 – May 2026*

Center for Theoretical Neuroscience, Columbia Zuckerman Institute *New York*

- Designed and implemented a four-direction spatial masking strategy for self-supervised denoising of two-photon calcium-imaging data, integrating it with PMD spatial-component selection and achieving up to 20% reduction in residual energy on an individual dataset.
- Developed and evaluated denoising configurations using ResNet, U-Net, and lightweight convolutional networks; conducted ablations of receptive fields, normalization, training settings, loss functions, and spatial-component selection.
- Studied Noise2Noise training-pair construction using shared versus independently estimated temporal bases, varying frame fractions, and independently corrupted observations to analyze component stability and limitations of self-supervised training.
- Constructed synthetic imaging datasets using real neuronal spatial footprints, simulated temporal traces, and controlled Gaussian noise; evaluated reconstruction quality and signal recovery across noise levels.

- Diagnosed a numerical discrepancy in PMD's blockwise spatial update and contributed a fixed-temporal-basis least-squares refitting correction through pull requests.
- Analyzed denoising at the spatial-component and reconstructed-video levels, comparing spatial, temporal, and combined denoising approaches, characterizing trade-offs in signal retention, temporal correlation, reconstruction fidelity, and rank selection.
- Organized laboratory research code into a downloadable software package and developed public documentation and step-by-step guides for wet-lab researchers.

Selected Projects & Activities

Hackathon Team Lead - 2nd Place, XRPL ($5K); 3rd Place, Stellar ($3K) *Sep 2025 – Oct 2025*

- Independently led the technical development of two award-winning 4-person hackathon teams, driving product architecture, implementation, integration, and debugging while coordinating team strategy and engagement with competition judges. See code.
- Built VenMorph, an XRPL-based payment platform integrating wallet, Flare price oracles, payment requests, and on-chain transaction verification; and a Stellar structured-finance platform featuring senior/junior tranches, cash-flow waterfalls, loss simulations, wallet integration, and Soroban smart-contract interactions.

FlyWire Connectome Analysis - Computational Statistics Project *May 2025*

- Implemented Bayesian stochastic block models with class-dependent and spatially informed connectivity likelihoods to infer latent neuronal types and class-dependent wiring patterns from FlyWire connectomes.
- Applied Bayesian connectivity models to FlyWire chemical-synapse networks, constructing thresholded adjacency matrices and evaluating cell-type recovery across brain regions; identified limitations of distance-dependent assumptions in heterogeneous neural circuits.
- Developed a burst-based Poisson spike-train simulator and incorporated co-firing-based functional connectivity into a joint structural–functional inference objective, evaluating latent-block recovery and unresolved class distinctions on simulated neural activity. See paper.

Columbia AI Alignment Club - Technical Fellow *Feb 2025 – May 2025*

- Reproduced and critically evaluated Anthropic-style **hallucination-circuit analysis** in Qwen3-4B using pretrained transcoders and feature-attribution graphs, investigating candidate refusal-related mechanisms and limitations of circuit faithfulness.
- Implemented activation hooks across 28 transformer layers, linear probes, and bidirectional steering scripts with randomized-direction, sham-intervention, and grouped-bootstrap controls; validated Qwen3-8B FP32 inference and Hugging Face–TransformerLens consistency, with logit discrepancies below 4.2×10^{-5}.
- Designed controlled known-versus-unknown entity experiments on Qwen3-8B, evaluating 40 natural-language responses across matched prompt templates and multiple random seeds; identified unstable answer-withholding behavior that prevented progression to reliable causal circuit analysis.
- Analyzed full-response semantics and token-level generation trajectories, identifying a shared-prefix case in which factual attribution and answer withholding diverged at the seventh generated token; demonstrated why first-token logit shifts cannot independently establish semantic hallucination effects.

Brain Computer Interface Lab - EEG Decoding Project *January 2025*

- Developed an EEG motor-imagery decoding pipeline with CSP-based features, training-only fitting, and run-grouped validation, improving offline accuracy from 67.90% to 77.39% over the CSP–LDA baseline.
- Implemented stateful causal filtering, ring buffers, and cue-triggered streaming inference; achieved 77.78% cross-day accuracy and exact streaming–batch agreement across 1,296 predictions.

Columbia Asian Faculty Association - Data Analyst *Jan 2025 – Mar 2025*

- Analyzed faculty survey data on workload, well-being, and institutional support under faculty supervision; presented findings in a formal report to board members for decision-making.

Technical Skills

Fluent in: Python, SQL, R **ML:** PyTorch, Hugging Face, scikit-learn
Tools: Linux, Git, Docker, FastAPI, vLLM, Neo4j